

# **SMART WAVE ELECTRONICS PROJECT**

**TITLE** : Smart Wave Control Switch with CD4017

## **WORKING PRINCIPLE :**

The **Smart Wave Control Switch** operates based on detecting **motion** using an **infrared (IR) sensor**, which then triggers a **CD4017 Decade Counter IC**. The IC, in turn, controls a **relay** or **LED** to turn **on** or **off** based on the motion detection.

### **1. IR Sensor (IR LED pair – Emitter & Detector):**

- The **IR emitter** emits **infrared light**.
- The **IR detector** detects the **reflected infrared light**. If there is **motion** in the path of the IR beam, the **reflection changes**, causing the detector to **send a signal**.

### **2. Signal Processing (LM358 and Transistor - BC547):**

- The **LM358 IC** is used as a **comparator** or **amplifier**. It **processes the signal** received from the **IR sensor**.
- The **output** of the LM358 is connected to the **BC547 transistor**, which acts as a **switch**.

- When the LM358 detects a change in the signal from the IR sensor (due to motion), it sends a **triggering signal** to the BC547.
- The **transistor** is used to **drive higher current components** (like a relay or LED), since the IR sensor and LM358 provide **small currents**.

### 3. CD4017 Decade Counter IC:

- The **CD4017 IC** is a **7-output decade counter** that increments its output on each **clock pulse**.
- The **output pins** can be connected to the **relay** or **LEDs**.
- On receiving a pulse from the **motion detection circuit**, it **activates outputs sequentially**.
- Depending on design, the CD4017 powers the relay or LEDs based on which **output is active**.

### 4. Relay & LED Output:

- When a specific **CD4017 output** is triggered, it energizes a **5V SPDT relay** to control an external device (like a **fan** or **light**) or **turns on LEDs**.
- If the output is connected to a **relay**, it switches the device **on/off**.
- If connected to **LEDs**, the CD4017 will **toggle them** based on the counter's output.

## 5. Capacitors and Resistors:

- **100 $\mu$ F** and **1000 $\mu$ F Capacitors** are used for **power smoothing** and **voltage stabilization**.
- The larger capacitor helps in **filtering noise** or **fluctuations** in the power supply.
- **Resistors (220 $\Omega$  and 10k $\Omega$ ):** Used for **current-limiting** and **biasing** the transistor and IC.
- The **10k $\Omega$  trimmer** helps **fine-tune** the sensor's **motion detection threshold**.

## 6. 1N4007 Diode:

- The **1N4007 diode** provides **flyback protection** across the **relay coil**, protecting the circuit from **voltage spikes** when the relay is **de-energized**.

## COMPONENTS USED :

- CD4017 IC
- LM358 IC
- BC547 Transistor
- 100 $\mu$ F 25V Capacitor
- 1000 $\mu$ F 25V Capacitor
- 220-ohm 0.25W Resistors
- 10k 0.25W Resistor

- 10k Trimmer
- LED 5mm
- IR LED pair (Detector & Emitter)
- 1N4007 Diode
- 5V SPDT Relay
- Connectors & IC Base
- PCB Board
- Soldering Iron

## CIRCUIT DIAGRAM :

